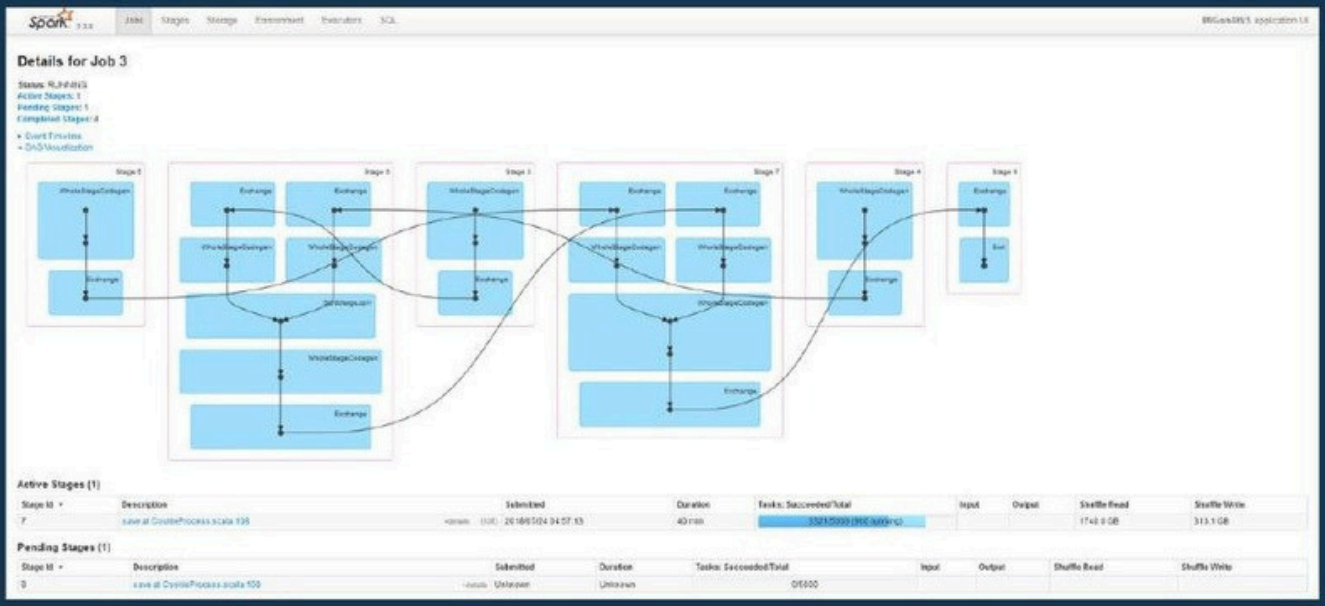
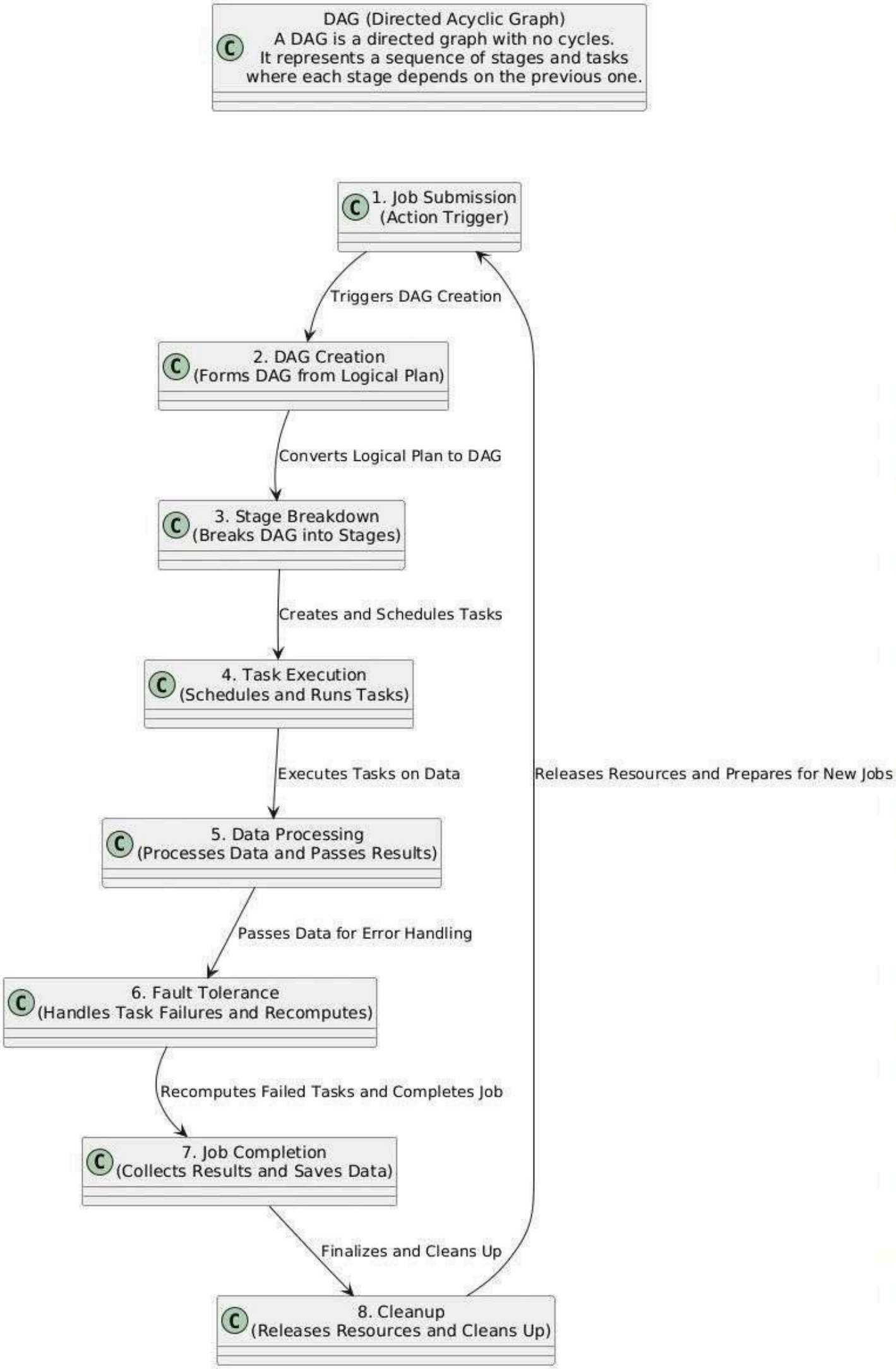
Apache Spark DAG Features

Directed Acyclic Graph - Brain of Spark Job



Spark DAG Operation



DAG in Apache Spark

Stage Pipelining
Multiple transformations can be grouped into a single stage for optimized execution

Fault Isolation
Only falled tasks/stages are recomputed, not the entire job

Lineage Tracking
Tracks every transformation for fault recovery without data duplication

Parallel Task Scheduling
Tasks run concurrently across the cluster for maximum speed

Deterministic Execution
Guarantees consistent results for the same input every time

Spark UI Visualization
DAGs can be tracked visually in the Spark UI for debugging and insights

Logical vs Physical Plan Separation
DAG is optimized from a logical plan before execution begins

Resource Optimization
Smart caching and reuse of computations for better cluster utilization

🔍 Advanced Features of DAG in Spark

1️⃣ Stage Pipelining

- Spark can **pipeline transformations** within the same stage to reduce execution time.
- Example: `map → filter → flatMap` can be executed together without shuffle.

2️⃣ Lineage Tracking

- Every RDD or DataFrame transformation maintains a **lineage graph**.
- Spark uses this to **recompute lost partitions** in case of failure—**no data duplication required**.

3️⃣ Deterministic Computation

- DAG ensures **determinism** in execution: the same inputs and transformations will **always** yield the same results.

4️⃣ Logical vs Physical Plan Separation

- DAG is derived from Spark's **logical plan**, which Spark then **optimizes** into a **physical plan** for execution.
- This allows **query optimization** (via Catalyst Optimizer) before execution starts.

5️⃣ Lazy Evaluation

- Transformations are not executed immediately—only **actions** trigger the DAG.
- This **reduces unnecessary computation** and allows Spark to optimize the entire plan first.

6 Fault Isolation

- In case of task failure, **only the dependent stage** is recomputed—not the entire pipeline.
- This minimizes recovery time and saves cluster resources.

7 Parallel Task Scheduling

- DAG allows **tasks within a stage** to run **in parallel** across multiple executors.
- This ensures **maximum resource utilization** and **faster job completion**.

8 Spark UI Visualization

- DAG is **visually available** in the Spark UI under the "Stages" tab.
- Developers can **track job execution**, bottlenecks, and failed stages for deep debugging.

9 Resource Optimization

- Spark uses the DAG to determine **where and when to cache** intermediate results or reuse computations.
- Helps avoid recomputation in iterative algorithms (like ML or GraphX).

10 Support for Complex Workflows

- DAG supports **complex dependencies**, allowing Spark to handle jobs with **multiple branches**, **joins**, **co-groups**, and more with efficiency.

🔍 Advanced Features of DAG in Spark

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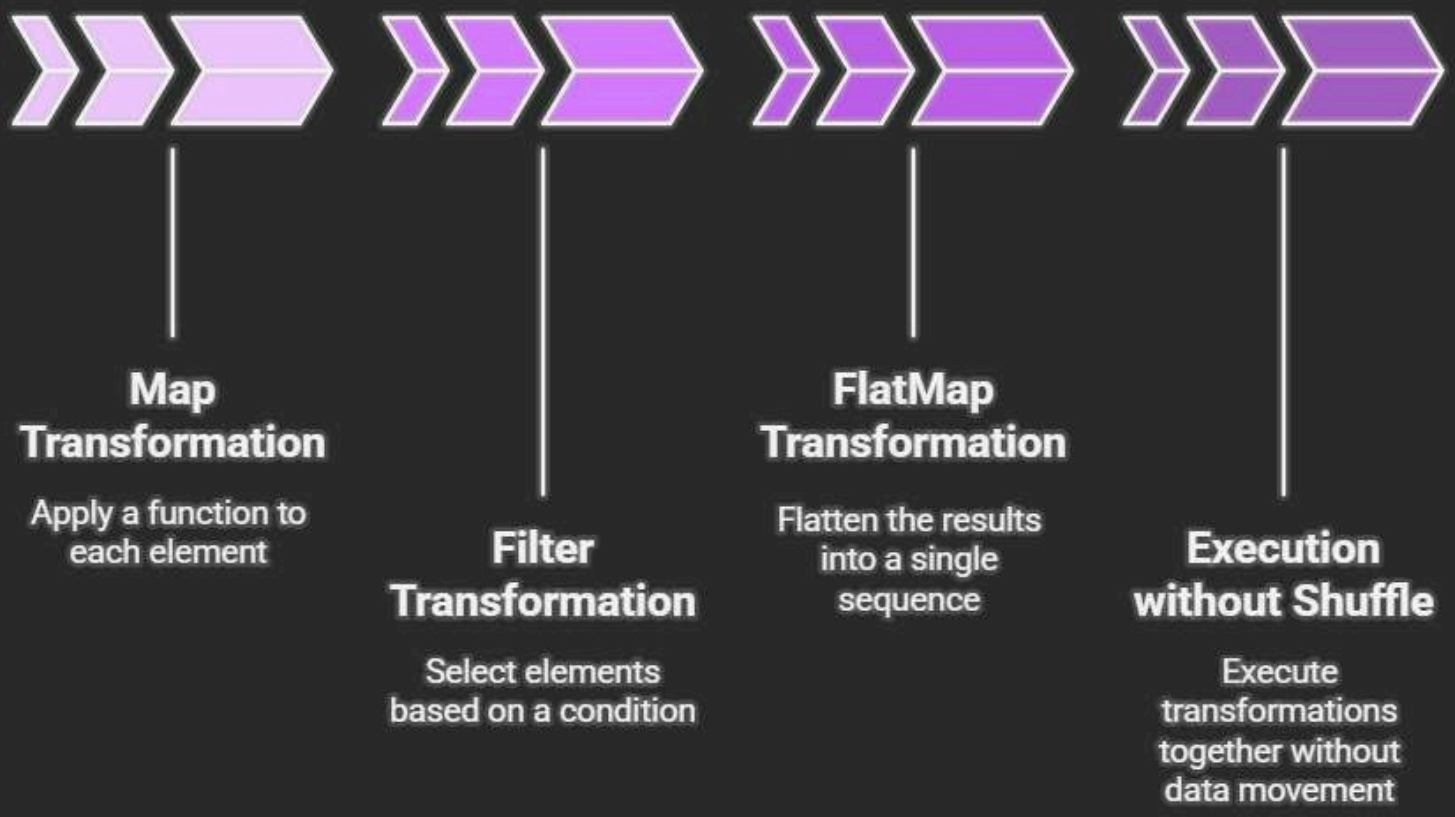

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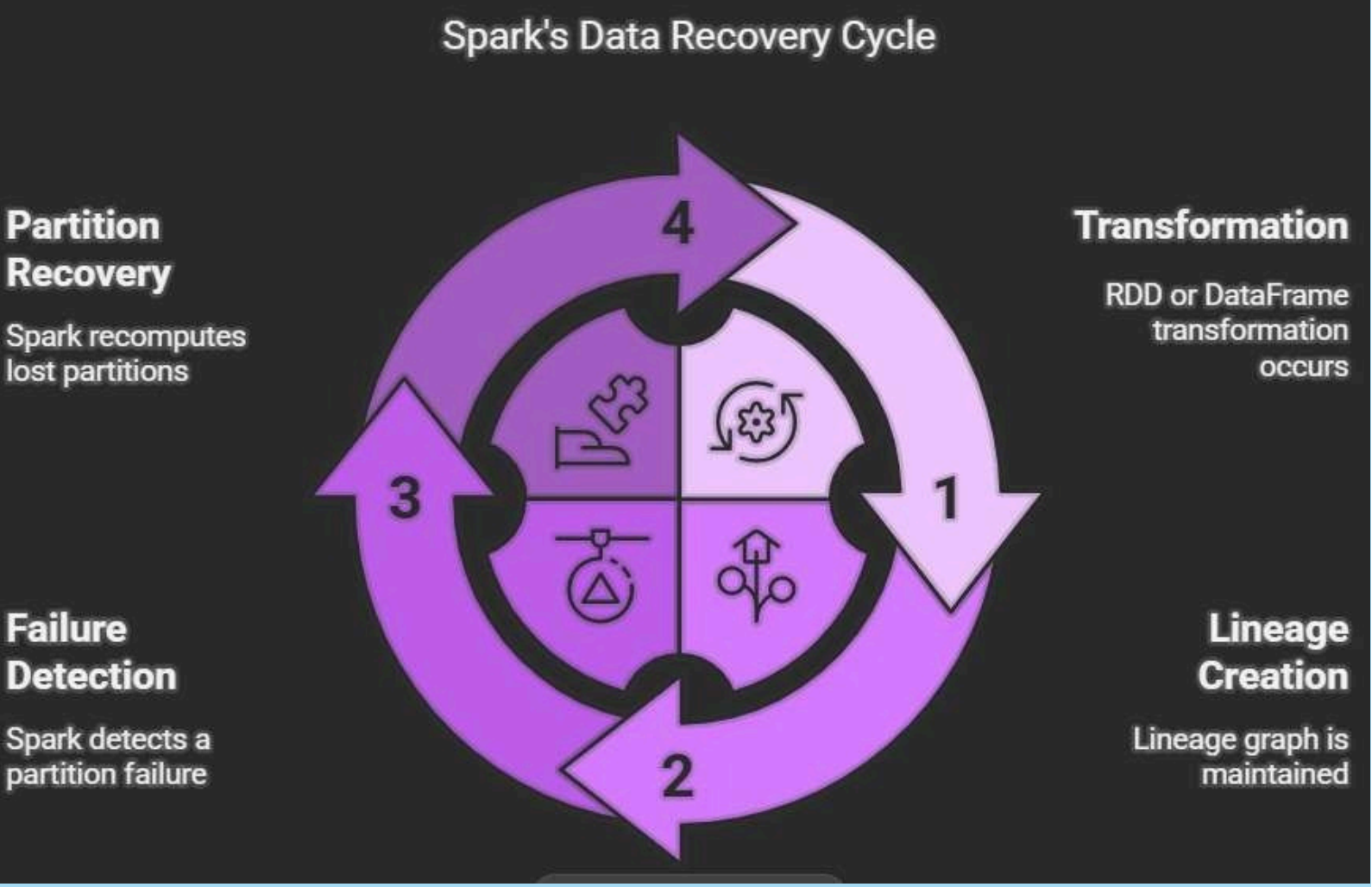

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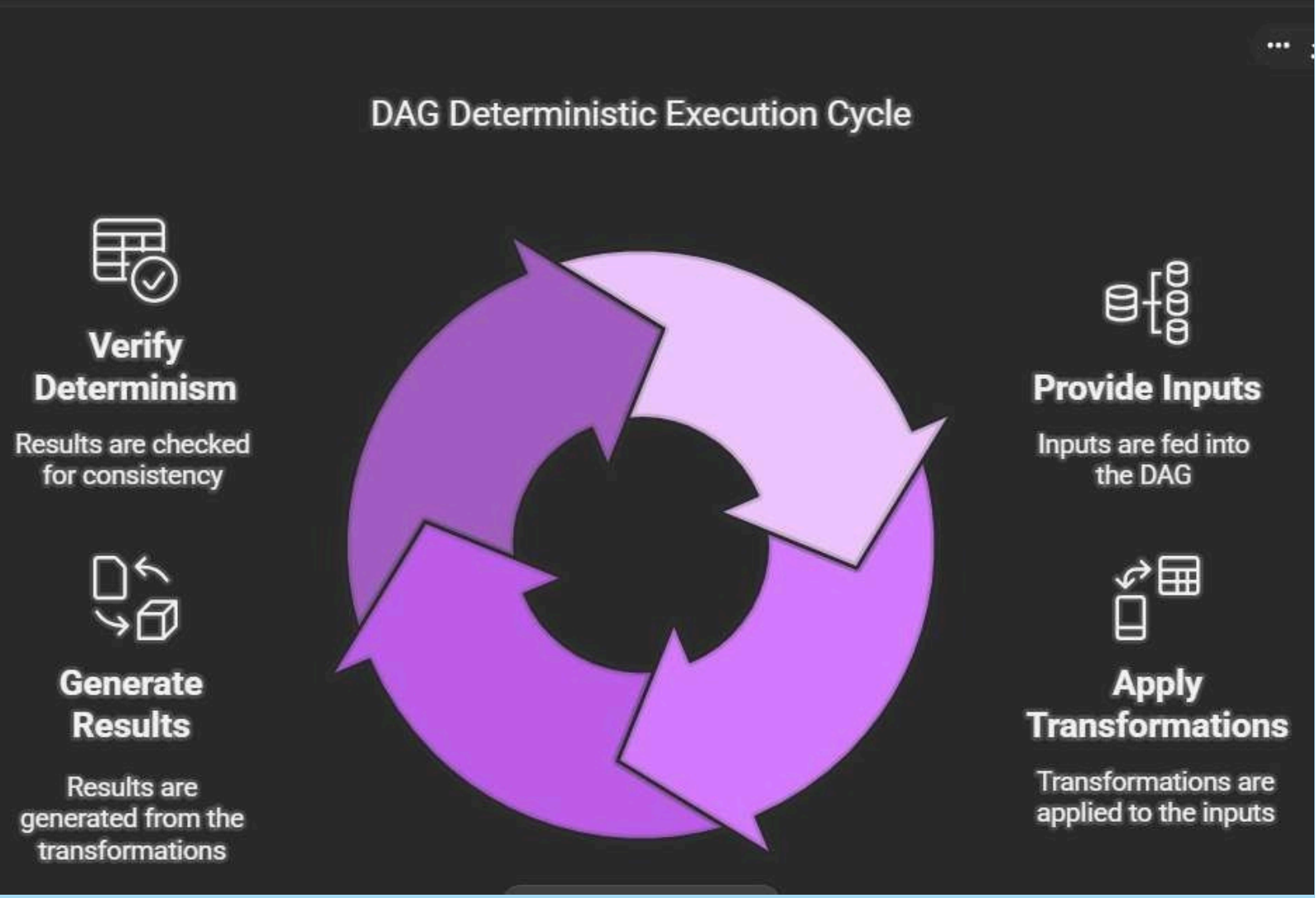

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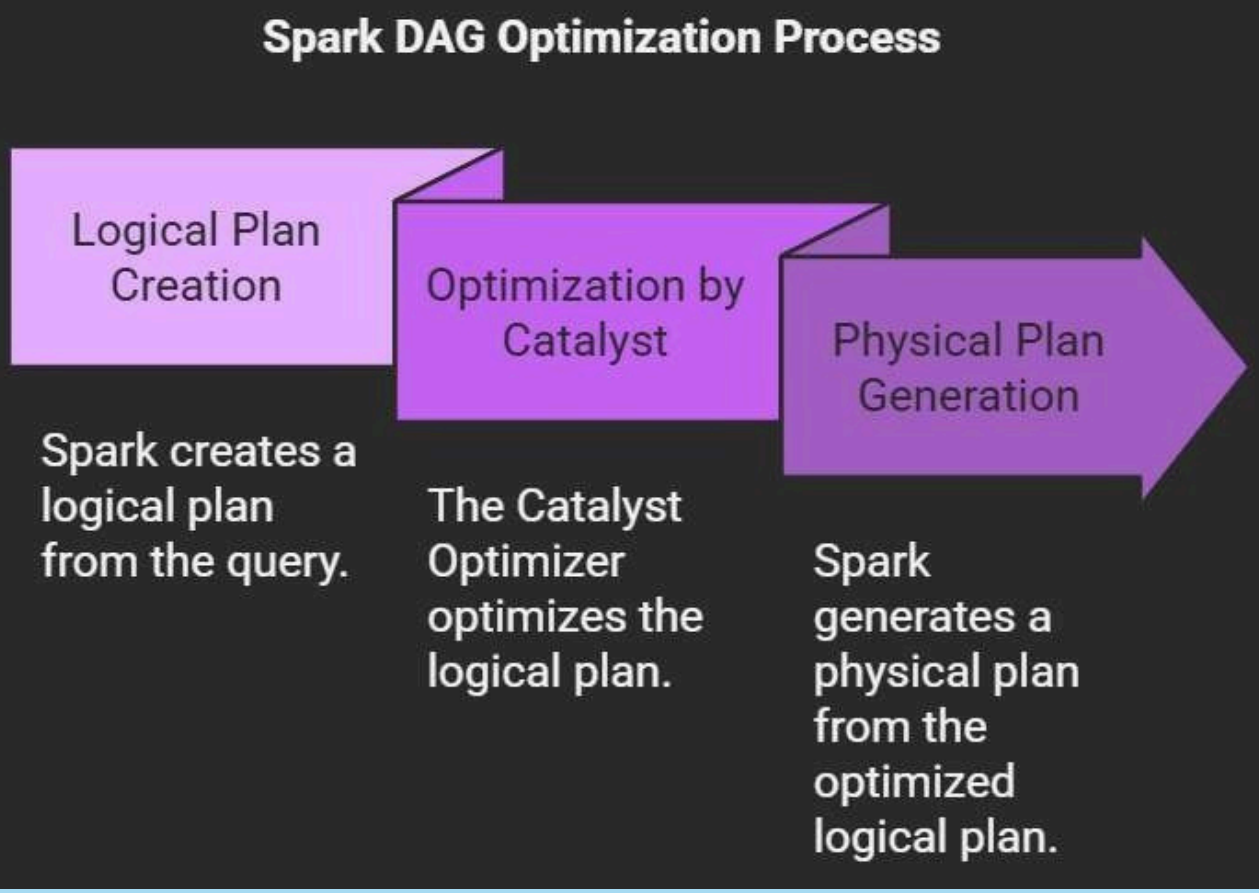

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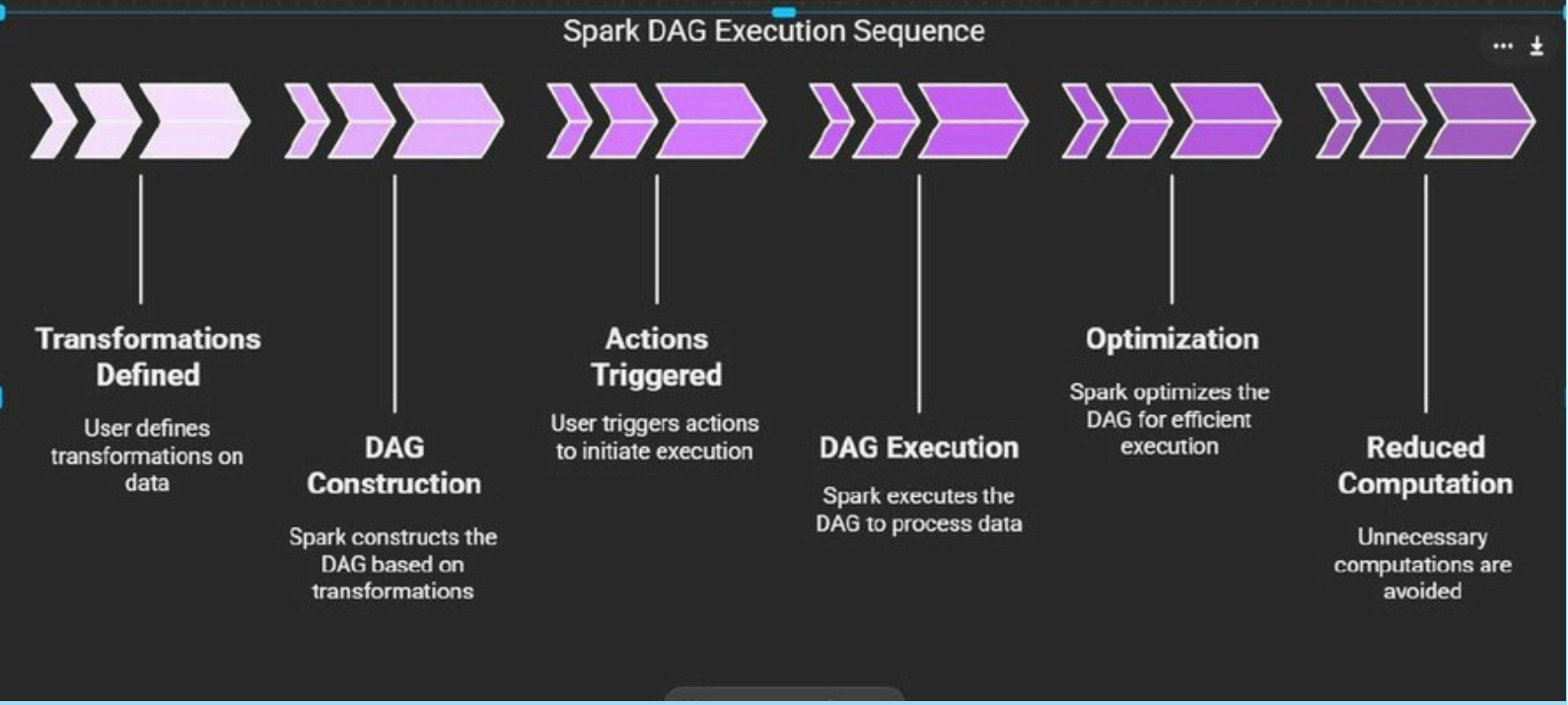

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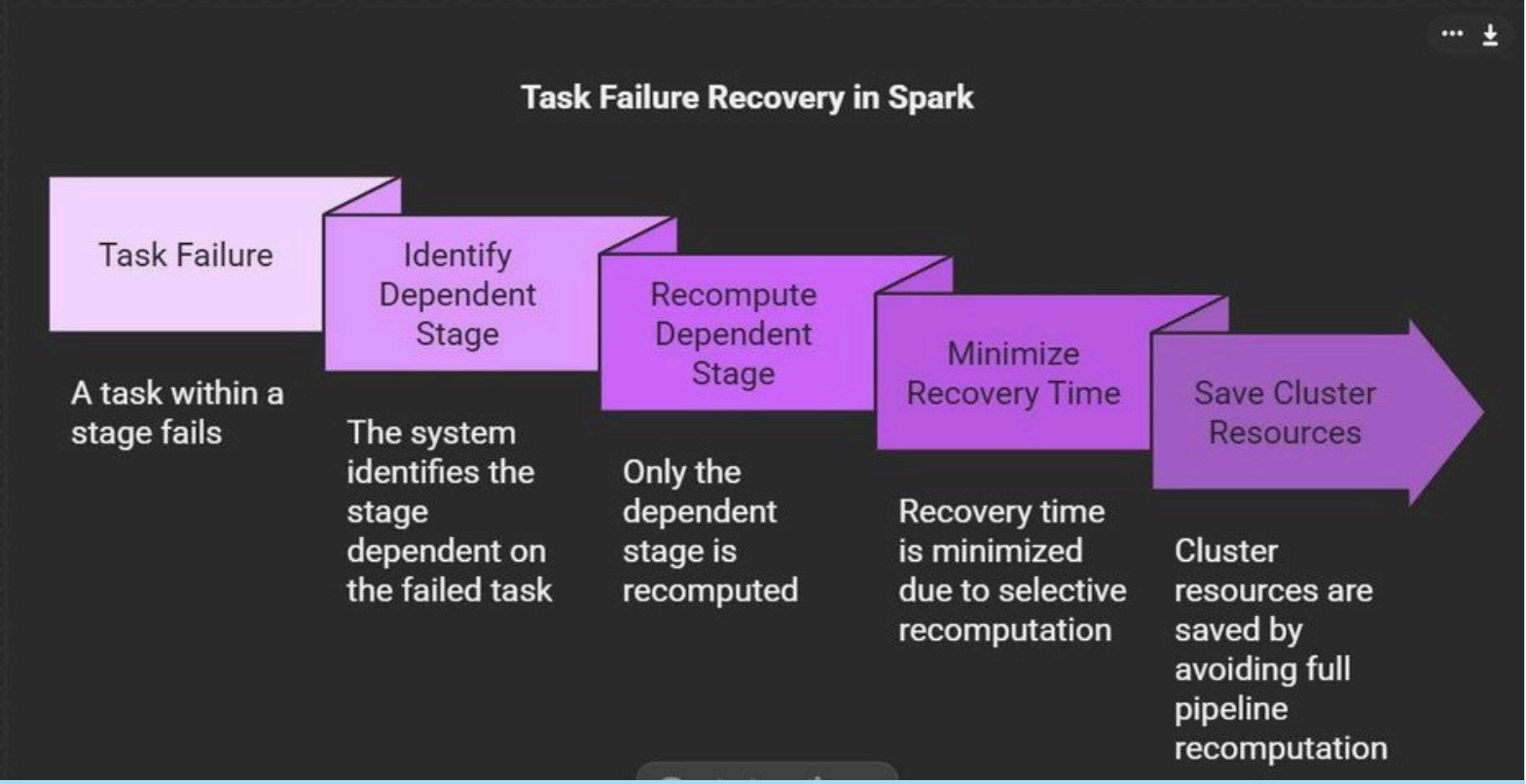

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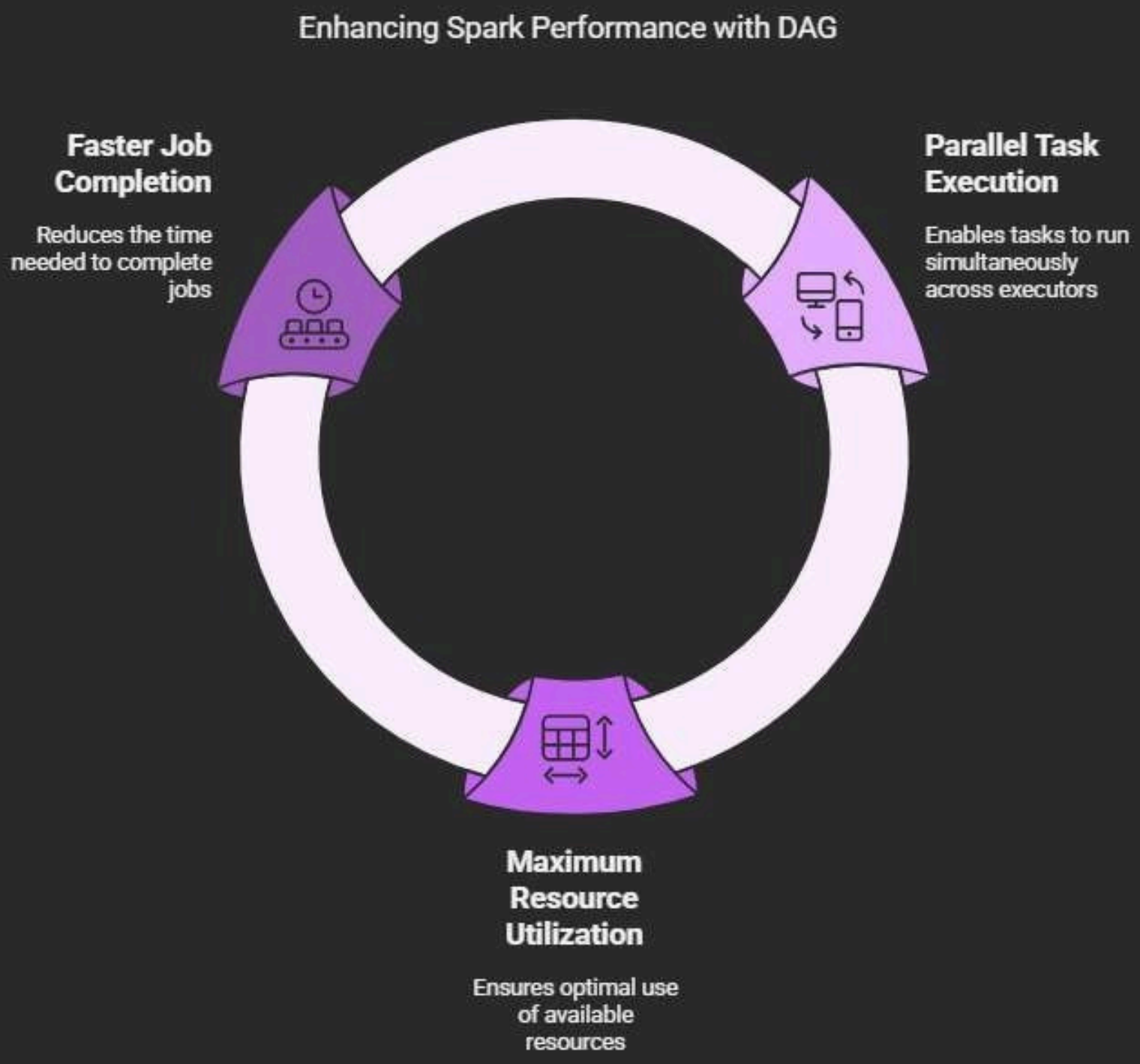

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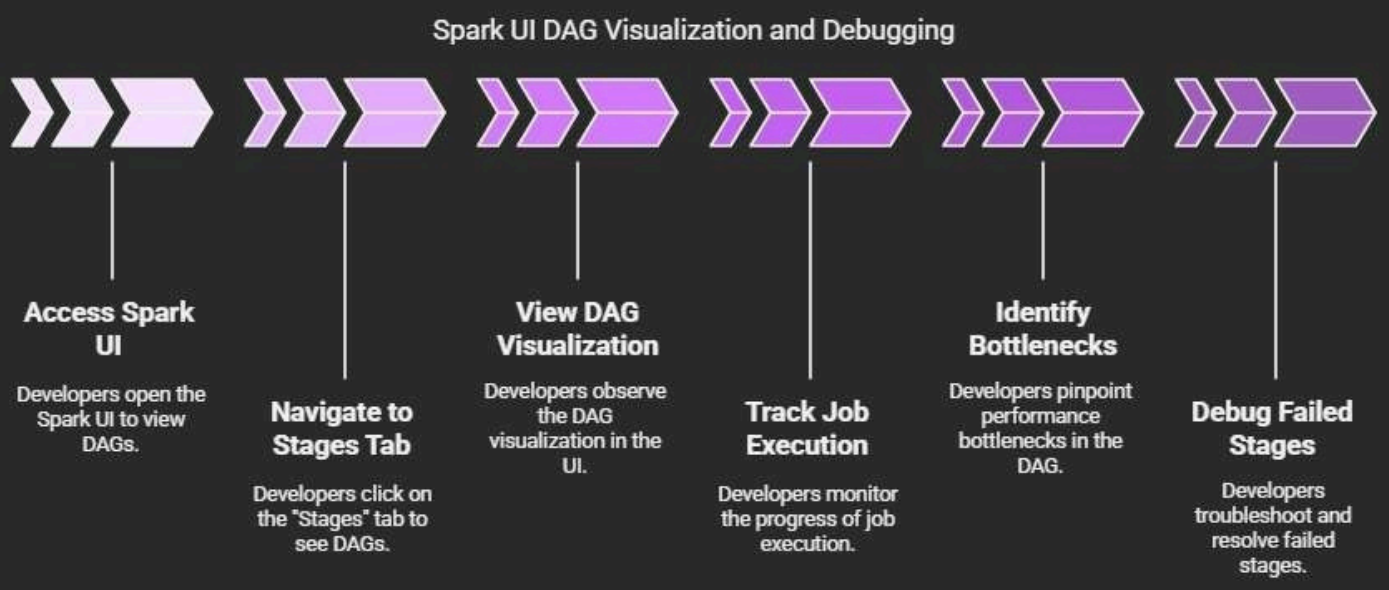

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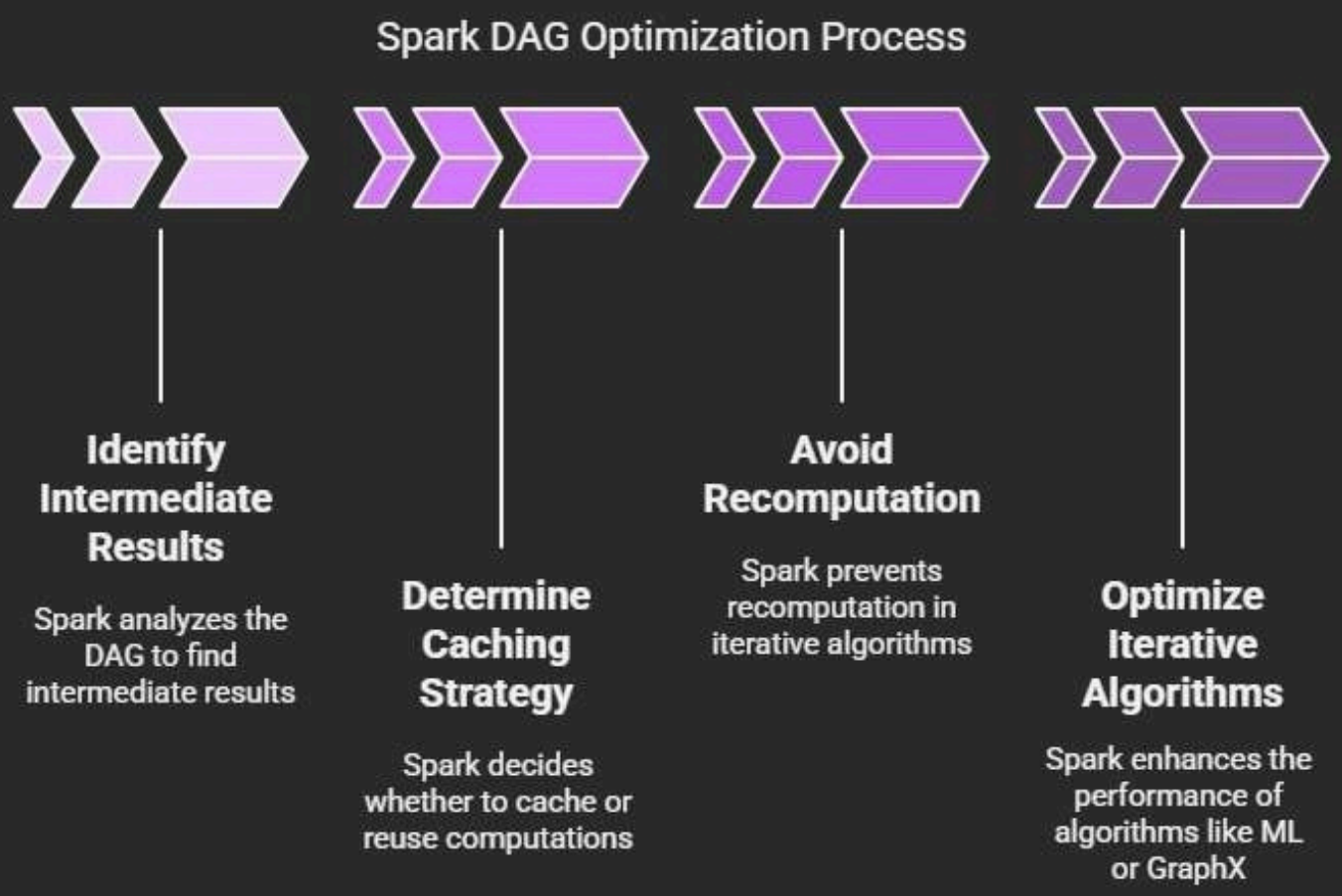

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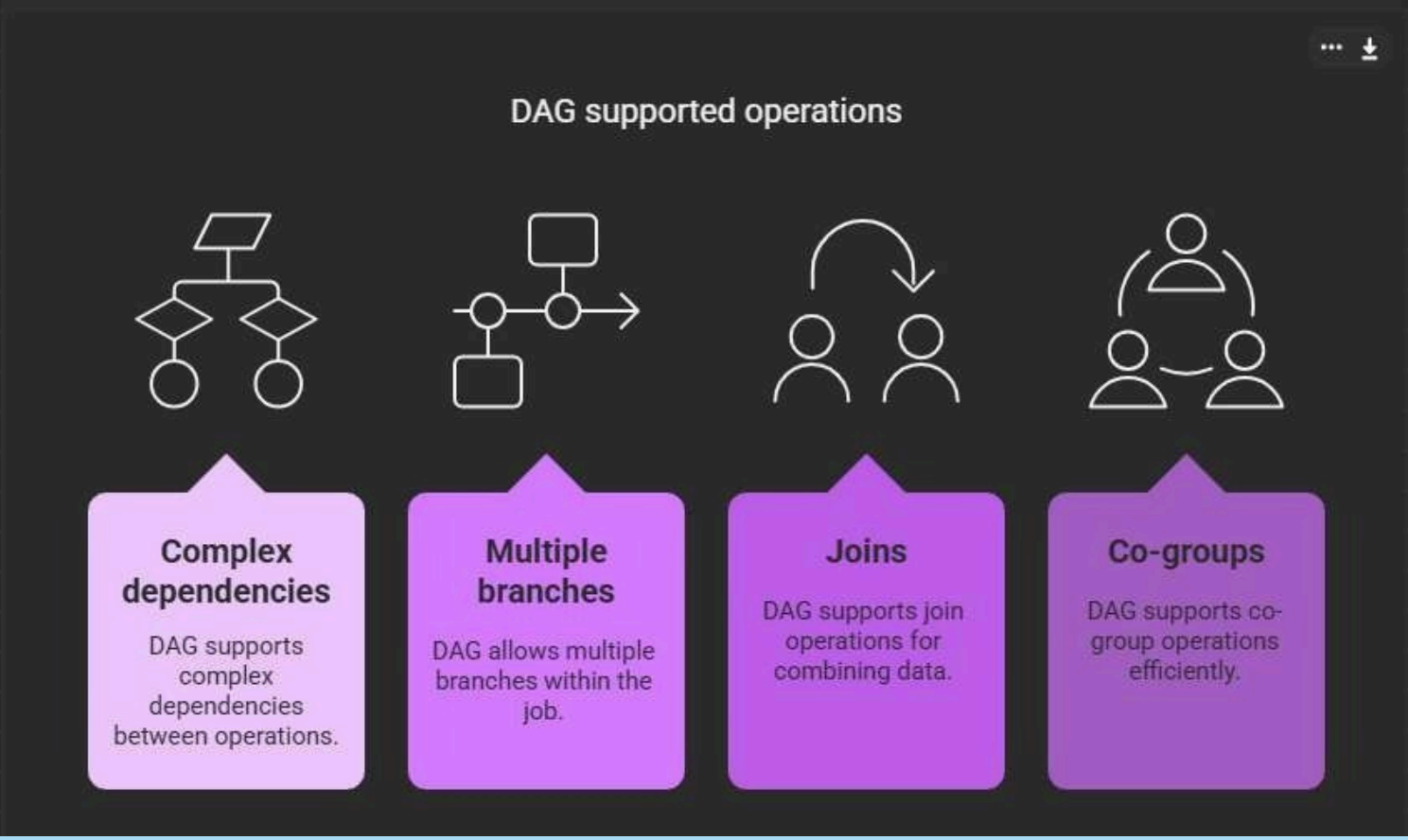
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